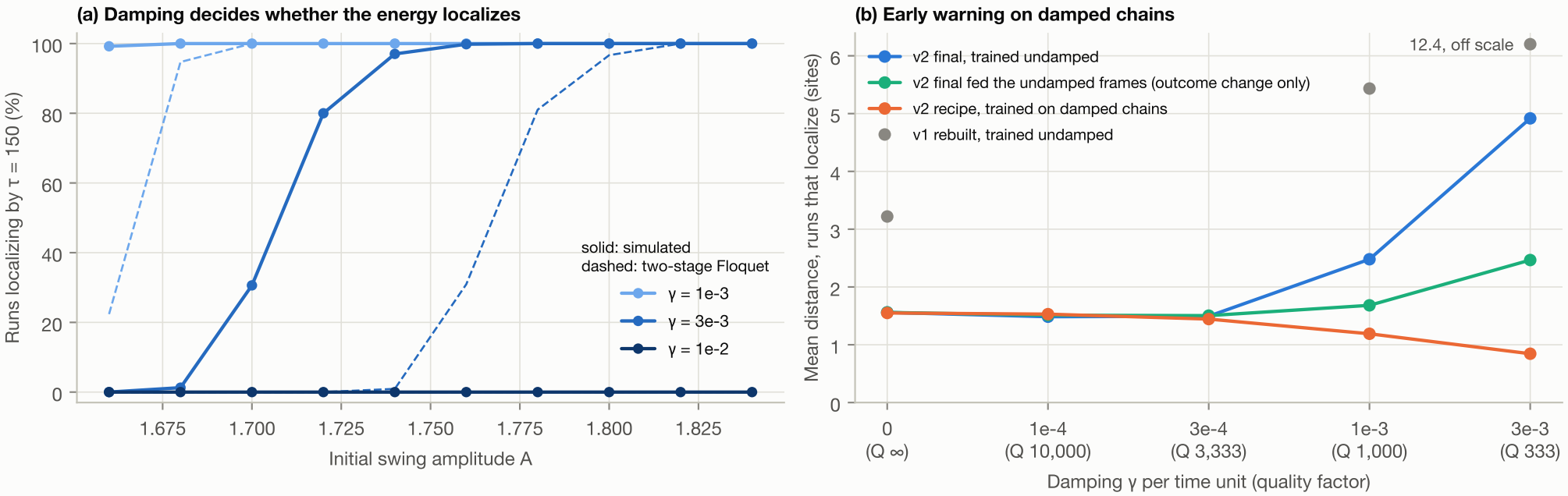
Linear damping: the swing decays toward A_c = 1.577 (5,000 runs per γ, shared initial conditions)
(a) Damping decides whether the energy localizes
Runs localizing by τ = 150 (%)
100
80
60
40
20
0
1.675
1.700
1.725
1.750
1.775
1.800
1.825
Initial swing amplitude A
solid: simulated
dashed: two-stage Floquet
γ = 1e-3
γ = 3e-3
γ = 1e-2
(b) Early warning on damped chains
Mean distance, runs that localize (sites)
6
5
4
3
2
1
0
12.4, off scale
v2 final, trained undamped
v2 final fed the undamped frames (outcome change only)
v2 recipe, trained on damped chains
v1 rebuilt, trained undamped
0
(Q ∞)
1e-4
(Q 10,000)
3e-4
(Q 3,333)
1e-3
(Q 1,000)
3e-3
(Q 333)
Damping γ per time unit (quality factor)